

Euclidean and non-Euclidean 3-dimensional spaces is a fundamental area of research in geometry and physics. Over time, researchers have developed various results, theorems, and insights concerning these surfaces. In [8], the authors investigated the Bertrand offset for ruled surfaces and showed that, similarly to planar curves, a ruled surface can have an infinite number of Bertrand offsets. The relationship between the Bertrand offsets of trajectory ruled surfaces and their projections on spherical areas has been explored in [9, 10], along with their corresponding invariants. The Bertrand offsets of ruled surfaces in Minkowski 3-space were considered in [11, 12]. Additionally, the study of Mannheim offsets for timelike ruled and developable surfaces, particularly their invariants, was addressed in [13, 14]. In [15], Senturk and Yuce analyzed involute-evolute offsets of ruled surfaces and developed associated invariants using the geodesic Frenet frame. More recently, research on evolute offsets of ruled surfaces, especially those with constant Gaussian and mean curvatures, in both Euclidean and Minkowski 3-spaces has been presented in [16, 17].

The characteristics of ruled surfaces and their offset surfaces have been extensively studied in both Euclidean and non-Euclidean spaces (see [15-21]). However, existing literature lacks a detailed approach to constructing evolute offsets of slant $\mathfrak{TL}$ -ruled surfaces in terms of the striction curve. In this study, we explore the geometric properties of slant $\mathfrak{TL}$ -ruled surfaces and their evolute offsets within three-dimensional Minkowski space $\mathbb{E}_1^3$. By establishing a bijective correspondence through their rulings, we derive conditions under which an evolute offset $\mathfrak{M}^*$ maintains a coaxial relationship with the central normal of $\mathfrak{M}$. Furthermore, we formulate expressions governing curvature behavior and classify $\mathfrak{M}$ and $\mathfrak{M}^*$ based on specific functional parameters. Special cases, including $\mathfrak{TL}$ -developable, $\mathfrak{TL}$ -binormal, and $\mathfrak{TL}$ -cone surfaces, are also examined, with graphical visualizations provided to support the analysis.

This paper investigates slant timelike ($\mathfrak{TL}$) ruled surfaces and their evolute offsets in Minkowski 3-space $\mathbb{E}_1^3$. We present a parametric formulation of skew $\mathfrak{TL}$ -ruled surfaces and establish conditions ensuring the coaxial alignment of the central normal with the ruling direction of the corresponding offset surface. The study explores the geometric properties using the Blaschke and Darboux frames, deriving curvature characteristics and fundamental invariants. Special cases, including $\mathfrak{TL}$ -developable and $\mathfrak{TL}$ -binormal surfaces, are analyzed with illustrative examples. The findings provide a deeper understanding of the differential geometry of $\mathfrak{TL}$ -ruled surfaces and their evolute offsets in Lorentzian space.

2 Basic concepts

Let $\mathbb{E}_1^3$ denote Minkowski 3-space [22, 23]. For vectors $\mathfrak{e} = (e_1, e_2, e_3)$ and $\nu = (\nu_1, \nu_2, \nu_3)$ in $\mathbb{E}_1^3$, the inner product is defined as:

$$\langle \mathfrak{e}, \nu \rangle = e_1\nu_1 + e_2\nu_2 - e_3\nu_3.$$